<u>**Ex2 – Affine Spaces and Modeling**</u>

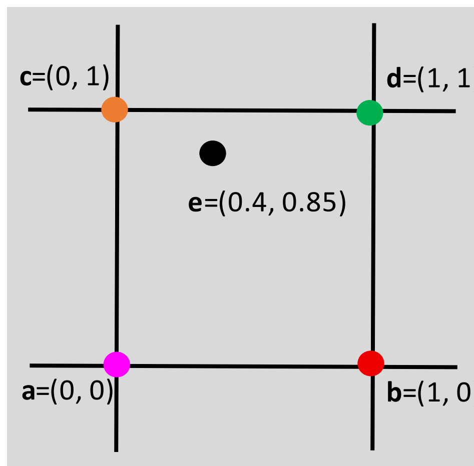
Please write your full name and ID at the beginning of your submission.

You must show and explain all your steps and calculations.

<u>**Question 1 (30 pts) – Image Processing and Affine Spaces**</u>

Each question stands by itself and is unrelated to the others.

1) (8 pts) Let **a, b, c, d, e** be five points on $\mathbb{R}^2$, defined as follows:

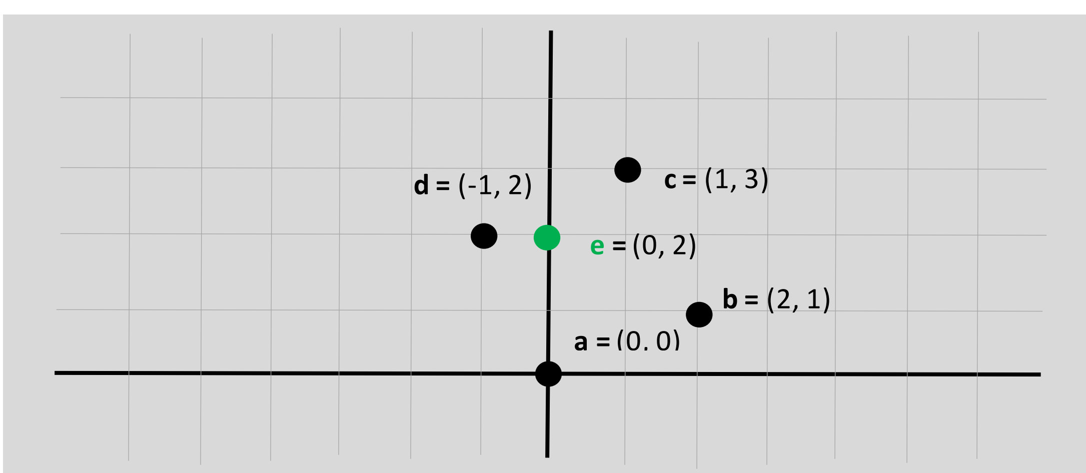


The RGB color values of **a, b, c, d** are:

a = (255, 0, 255), b = (0, 0, 255), c = (255, 165, 0), d = (0, 255, 0)

Calculate the RGB value of point e using bi-linear interpolation.

2) (12 pts) Let **a, b, c, d, e** be five points on $\mathbb{R}^2$, defined as follows:

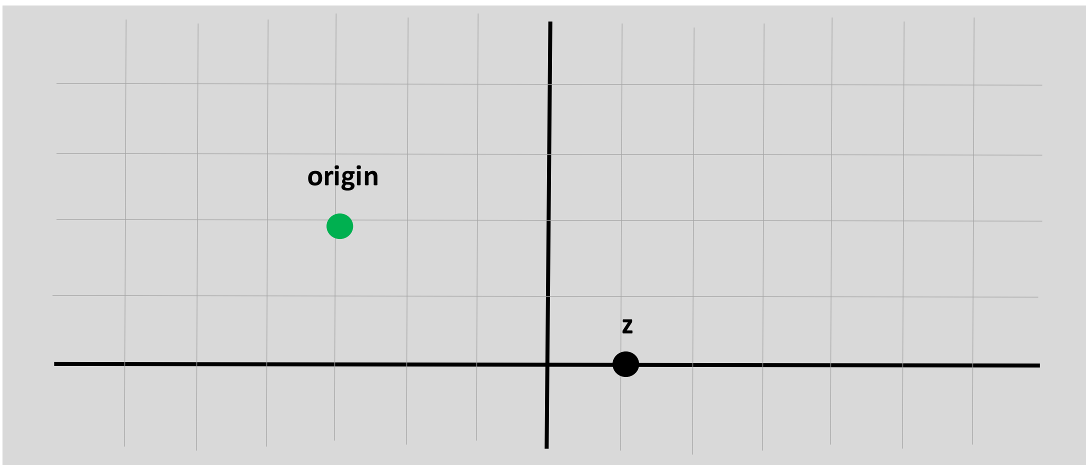


Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined for **a, b, c, d** as follows:

$f(a) = 3, f(b) = 1, f(c) = 21, f(d) = 8$

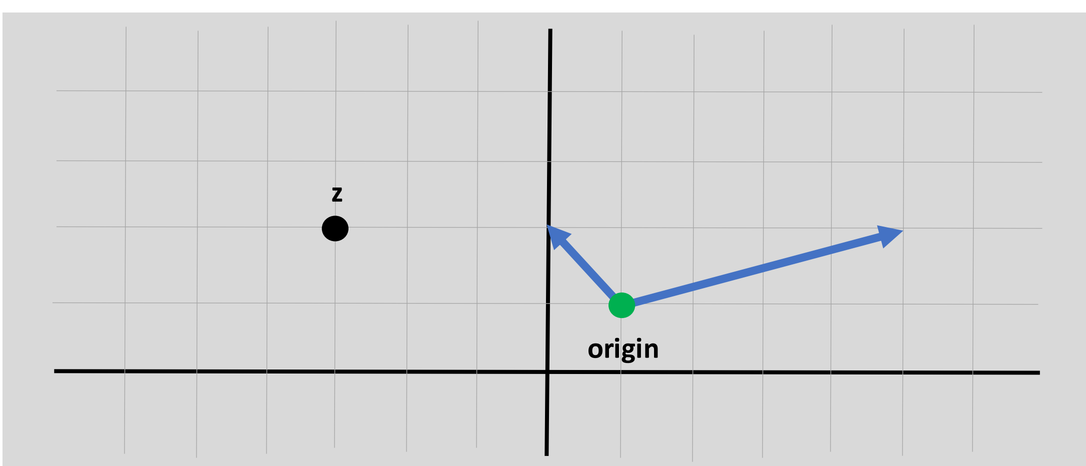
Calculate $f(e)$ using bi-linear interpolation.

3) Given the following affine space defined by the origin: (-3, 2)



a) (2.5 pts) Draw the vector (-1, -2) on the coordinate system (starting from the origin).
b) (2.5 pts) Find the (x, y) coordinates of point **z.**

4) Given the following affine space defined by the origin point (1, 1) and base vectors: $v_1 = (-1, 1)$ $v2 = (4, 1)$



a) (2.5 pts) Draw the point (-4, -0.5) on the coordinate system.
b) (2.5 pts) Find the (x, y) coordinates of **z**.

Question 2 (70 pts) – Geometry and Modeling

5) Let $l = (1,4,-3) + t(2,2,2)$ be a line in E^3 (the Euclidean 3d space)
   a) (5 pts) Let $\boldsymbol{a} = (\mathbf{4}, -\mathbf{1}, \mathbf{4})$. Calculate the projection of point $\boldsymbol{a}$ on l and its distance from the line.
   b) (6 pts) let $l_2 = (2,3,2) + s(0, y, -3)$. Let $y_1 = -3, y_2 = 2$. Check for both y_1, y_2 if l_2 intersects l. if the lines intersect, find the intersection point and the angle between lines. Otherwise, calculate the distance between the lines
      - **<u>Reminder</u>: The distance between two lines is defined as the smallest distance between two points on $\boldsymbol{l}$ and $\boldsymbol{l_2}$.**

6) Let $\pi_1 : -x + 4y + 2z = 1$ be a plane and $p = (5,-5,1)$ a point.
   a) (4 pts) Define the line equation of the line l which is perpendicular to the π_1 and passes through p.
   b) (6 pts) Define the sphere equation which is centered at p and is tangent to the plane.
   c) (4 pts) Find the tangent point of the sphere and the plane.
   d) (8 pts) Let $\pi_2 : x + 3 - 4z = 3$ be a plane. Find the parametric line representation of the intersection of the two planes and define a general equation for all spheres which are centered somewhere on the line and has p on their surface.

7) Let $(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 = 0$ be a sphere.
   a) (6 pts) It is known that every sphere can be enclosed by a cube. Define six plane equations which enclose the above sphere, and each is parallel to at least one axis $(x, y, or\ z)$.

b) (6 pts) It is known that every cube can be enclosed by a sphere. Calculate the equation of the minimal sphere enclosing the cube from (a).

8) Let $P_1 = (0,0,1), P_2 = (1,0,0), P_3 = (0,0,-1), P_4 = (-1,0,0)$
$P_5 = (0,\sqrt{2},1), P_6 = (1,\sqrt{2},0), P_7 = (0,\sqrt{2},-1), P_8 = (-1,\ \sqrt{2},0)$.

The following points define a cube in $\mathbb{R}^3$ with the following planes:

$$\begin{array}{ll} \pi_1: P_1, P_2, P_3, P_4 & \pi_4: P_2, P_3, P_6, P_7 \\ \pi_2: P_5, P_6, P_7, P_8 & \pi_5: P_1, P_2, P_5, P_6 \\ \pi_3: P_1, P_4, P_5, P_8 & \pi_6: P_3, P_4, P_7, P_8 \end{array}$$

(**See scene render on next page**)

a) (5 pts) Calculate the **outward** facing normal vectors of π_2, π_4, π_5, and deduce from these vectors the outward facing normal vectors of π_1, π_3, π_6. Use the scene render in the following page and the right-hand rule to determine to direction of the vector (even if you know how to do it without it).
b) (6 pts) Find the implicit representation of $\pi_1, \dots, \pi_6$.
c) (8 pts) Find the sphere equation for the sphere enclosed inside the cube.
d) (6 pts) For each of the following points in $\mathbb{R}^3$ classify if it resides (i) inside the cube and the sphere, (ii) inside the cube only, or (iii) outside of both.
   1) (0.5, 0.5, 0.6)
   2) (0.1, 0.3, 0.8)
   3) (0.1, 1.2, 0.2)

Scene Render